Practical No : 4

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Q.4) Write a C program for Merge Sort.

```
#include <stdio.h>
void merge(int arr[], int low, int mid, int high) {
    int i = low;
    int j = mid + 1;
    int k = 0;
    int temp[high - low + 1];
    while (i <= mid && j <= high) {
        if (arr[i] <= arr[j]) {
            temp[k++] = arr[i++];
        } else {
            temp[k++] = arr[j++];
        }
    }
    while (i <= mid) {
        temp[k++] = arr[i++];
    }
    while (j <= high) {
        temp[k++] = arr[j++];
    }
    for (i = low, k = 0; i <= high; i++, k++) {
        arr[i] = temp[k];
    }
}
void mergeSort(int arr[], int low, int high) {
    if (low < high) {
        int mid = low + (high - low) / 2;
```

```
    mergeSort(arr, low, mid);
        mergeSort(arr, mid + 1, high);
   merge(arr, low, mid, high);
    }
}
int main() {
    int n;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d elements: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    mergeSort(arr, 0, n - 1);
    printf("Sorted array: ");
    for (int i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
    return 0;
}
```

Output:

